
O-RAN Test and Integration Focus Group

Certification and Badging Profiles

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Contents

Foreword.....	3
Modal verbs terminology	3
1 Scope.....	4
2 References	4
2.1 Normative references	4
2.2 Informative references	4
3 Definition of terms, symbols and abbreviations	4
3.1 Terms	4
3.2 Symbols.....	5
3.3 Abbreviations	5
4 Introduction	5
4.1 Scope.....	5
5 Certification and Badging Profiles.....	5
5.1 O-RAN Certificates	5
5.1.1 A1 services on A1 interface of Near-RT RIC	5
5.1.2 A1 services on A1 interface of Non-RT RIC.....	6
5.2 O-RAN IOT Badges	6
5.2.1 A1 services on A1 interface	6
5.3 O-RAN E2E Badges	7
Annex (informative): Change history/Change request (history)	8

Foreword

This Technical Specification (TS) has been produced by TIFG (Test and Integration Focus Group) of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

- xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.
- yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.
- zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies Certification and Badging Profiles (CB-Profiles) for the purpose of O-RAN Certification and Badging.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

- [1] O-RAN.TIFG.TS.Cert-Badge.0: “Certification and Badging Processes and Procedures”
- [2] O-RAN.WG2.TS.A1AP: “A1 interface: Application Protocol”

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

Not applicable

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms [given in [i.1] and the following] apply:

Certification and Badging ICS Proforma (CB-ICS Proforma): An overview of all mandatory/optional/ conditional functions and capabilities defined in base specification(s)/profile and, if exists, in O-RAN Certification and Badging Profile. It helps vendors ensure that their products meet the necessary requirements for O-RAN Certificates and Badges. When the implementation is done, vendor declares which functions and capabilities have been implemented/supported, and so it becomes Certification and Badging ICS.

Certification and Badging ICS (CB-ICS): A statement made by the vendor of an implementation stating which functions and capabilities have been implemented/supported for the purpose of O-RAN Certificate or Badge. The Certification and Badging ICS (CB-ICS) is attached to O-RAN Certificate, IOT Badge or E2E Badge.

Certification and Badging Profile (CB-Profile): For the purpose of O-RAN Certification and Badging, the CB-Profile may constrain the existing options in base specification(s)/profile, and shall not impose requirements that conflict with the existing requirements in base specification(s)/profile (e.g. a function/capability/ parameter mandatory in the base specification shall not be optional in the profile; or adding new function/capability/parameter that does not exist in the base specification is not allowed).

3.2 Symbols

For the purposes of the present document, the symbols [given in [i.1] and the following] apply:

Void

3.3 Abbreviations

For the purposes of the present document, the abbreviations [given in [i.1] and the following] apply:

CB	Certificaiton and Badging
CB-ICS	Certification and Badging Implementation and Conformance Statement
CB-Profile	Certification and Badging Profile
ICS	Implementation Conformance Statement
TIFG	O-RAN Test and Integration Focus Group

4 Introduction

4.1 Scope

For the purpose of O-RAN Certification and Badging and to facilitate O-RAN productization, the O-RAN Certification and Badging Profile (CB-Profile) [1] may be used to constrain the existing options in base specification(s)/profile, but shall not impose requirements that conflict with the existing requirements in base specification(s)/profile (e.g. a function/capability/parameter mandatory in the base specification shall not be optional in the profile; or adding new function/capability/parameter that does not exist in the base specification is not allowed.)

For O-RAN certificates, one CB-Profile is defined for each device and interface/service under test, e.g. Open FH O-RU, Open FH O-DU. For O-RAN IOT Badge, one CB-Profile is defined for each interface/service under test including both devices in the pair, e.g. Open FH (including both O-RU and O-DU). For O-RAN E2E Badge, one CB-Profile is defined for the whole system under test (not to individual devices and interfaces) and to E2E functionalities and capabilities.

5 Certification and Badging Profiles

5.1 O-RAN Certificates

5.1.1 A1 services on A1 interface of Near-RT RIC

The O-RAN Certificate for A1 services of Near-RT RIC is defined in [1], clause 6.5. The services and related service procedures of A1 application protocol are specified in [2].

For the purpose of O-RAN Certificaiton of A1 services of Near-RT RIC, the following constraints applied:

- 1) At least one A1 service shall be implemented and supported by Near-RT RIC
- 2) If A1 policy management service (A1-P) is supported by Near-RT RIC (A1-P Producer), then the following A1 service procedures shall be supported:
 - a) Create Policy [2], clause 5.2.4.3

- b) Delete Policy [2], clause 5.2.4.6
- 3) If A1 enrichment information service (A1-EI) is supported by Near-RT RIC (A1-EI Consumer), then the following A1 service procedures shall be supported:
 - a) Create EI job [2], clause 5.3.4.3
 - b) Delete EI job [2], clause 5.3.4.6
 - c) Deliver EI job result [2], clause 5.3.5.2

5.1.2 A1 services on A1 interface of Non-RT RIC

The O-RAN Certificate for A1 services of Non-RT RIC is defined in [1], clause 6.4. The services and related service procedures of A1 application protocol are specified in [2].

For the purpose of O-RAN Certificaiton of A1 services of Non-RT RIC, the following constraints applied:

- 1) At least one A1 service shall be implemented and supported by Non-RT RIC
- 2) If A1 policy management service (A1-P) is supported by Non-RT RIC (A1-P Consumer), then the following A1 service procedures shall be supported:
 - a) Create policy [2], clause 5.2.4.3
 - b) Delete policy [2], clause 5.2.4.6
- 3) If A1 enrichment information service (A1-EI) is supported by Non-RT RIC (A1-EI Producer), then the following A1 service procedures shall be supported:
 - a) Create EI job [2], clause 5.3.4.3
 - b) Delete EI job [2], clause 5.3.4.6
 - c) Deliver EI job result [2], clause 5.3.5.2

5.2 O-RAN IOT Badges

5.2.1 A1 services on A1 interface

The O-RAN IOT Badge for A1 services between Near-RT RIC and Non-RT RIC is defined in [1], clause 7.6. The services and related service procedures of A1 application protocol are specified in [2].

For the purpose of O-RAN IOT Badge of A1 services of between Near-RT RIC and Non-RT RIC, the following constraints applied:

- 1) At least one A1 service shall be implemented and supported by Near-RT RIC and Non-RT RIC. Both Near-RT RIC and Non-RT RIC shall support and implement the same A1 service(s).
- 2) If A1 policy management service (A1-P) is supported by Near-RT RIC (A1-P Producer) and Non-RT RIC (A1-P Consumer), then the following A1 service procedures shall be supported:
 - a) Create policy [2], clause 5.2.4.3
 - b) Delete policy [2], clause 5.2.4.6
- 3) If A1 enrichment information service (A1-EI) is supported by Near-RT RIC (A1-EI Consumer) and Non-RT RIC (A1-EI Producer), then the following A1 service procedures shall be supported:
 - a) Create EI job [2], clause 5.3.4.3
 - b) Delete EI job [2], clause 5.3.4.6
 - c) Deliver EI job result [2], clause 5.3.5.2

5.3 O-RAN E2E Badges

Annex (informative): Change history/Change request (history)

Date	Revision	Description
2025.07.18	12.00	First version (v12.00)
2025.11.28	13.00	